

MOHAMED ASHISH ROSHAN

Software Developer | Backend Engineer

📍 Tiruchirappalli, Tamil Nadu | ✉️ mohamedashishroshanpc@gmail.com | 📞 +91-8148199506
🌐 [a roshan ash](#) | 📺 [AshishRoshan](#)

EXECUTIVE SUMMARY

Motivated and detail-oriented Software Engineer with a strong foundation in Python, Django, and FastAPI, capable of building scalable REST APIs and database-driven applications aligned with enterprise and consulting demands. Currently advancing skills in DevOps, with a focus on CI/CD pipelines, cloud infrastructure, and AWS EC2 deployments — bringing both development and operational thinking to every solution. Demonstrates solid knowledge across backend development, networking, hardware systems, and application support, with a practical ability to debug issues, analyze logs, and resolve problems efficiently — qualities essential in fast-paced IT staffing and consulting environments. Familiar with Git workflows, test-driven development using pytest and unittest, and cloud-based deployments — committed to delivering secure, scalable, and production-ready software that meets modern engineering and client-facing operational standards.

EDUCATION

B.E. Computer Science and Engineering – M.I.E.T Engineering College, Tiruchirappalli (CGPA: 8.5/10)	Sept 2022 – May 2026
HSC – Alpha Wisdom Vidyashram Senior Secondary School (70%)	2022
SSLC – Alpha Wisdom Vidyashram Senior Secondary School (76%)	2020

TECHNICAL SKILLS

Programming & Backend Development: Python, Django, FastAPI, REST API Development, SQL, Node.js **Frontend Development:** JavaScript, React.js, Bootstrap

Databases: PostgreSQL, MySQL, MongoDB, SQLite | Database Design, Query Optimization (Basic)

Application Support (Fundamentals): Basic knowledge of L1/L2 workflows, debugging, log analysis, and issue tracking

Networking & Systems: Computer Networks, Hardware Fundamentals, TCP/IP, OSI Model, System Troubleshooting

Hardware & Field Support: Hardware Assembly, Network Setup & Troubleshooting, CCTV Installation & Configuration, Structured Cabling

DevOps: Docker, GitHub Actions, CI/CD Pipelines, AWS (EC2 – Live Deployment, S3, RDS)

Testing & Code Quality: pytest, unittest, Unit Testing, Integration Testing, Test Coverage, Code Review

DSA & Problem Solving: Data Structures & Algorithms, LeetCode (Arrays, Strings, HashMaps, Dynamic Programming), Git Feature-Branch Workflow, Pull Requests

KEY PROJECTS

SentinelX AURAPRIME – DevSecOps Intelligence Platform Jan–Mar 2026
Team Lead & Full-Stack Developer

- Architected a DevSecOps-based platform integrating multiple security checks into a single system using **FastAPI** and **React**
- Engineered and managed **50+ REST APIs** for handling security data, analysis, and reporting, streamlining security workflows
- Designed a risk scoring system to visualize application security status through interactive dashboards
- Implemented automated vulnerability detection and patch validation workflow, reducing manual review effort significantly
- Integrated **Docker** for container scanning and strengthened CI/CD pipeline security
- Authored unit and integration tests using **pytest**, ensuring reliable test coverage across all API endpoints
- Led team using **feature-branch Git workflow** with pull requests and code reviews, maintaining code quality across the codebase

TrafficGuard – Web Application Firewall System

- Developed a web security system to filter and analyze incoming HTTP requests
- Implemented detection for common attacks such as **SQL Injection, XSS, and path traversal**
- Built rate-limiting and request monitoring features to reduce abnormal traffic
- Designed APIs using **Django REST Framework** for rule management and logging
- Created a simple dashboard to monitor traffic and security alerts

SysMonitor Pro – System Monitoring Platform

- Built a system monitoring tool to track application health and performance across services
- Implemented alerting and logging features for real-time incident tracking and resolution
- Used **Django REST** and **FastAPI** for backend services and APIs

- Developed real-time dashboards for visualizing system metrics and performance trends
- Containerized the application using **Docker** and **deployed live on AWS EC2**, making it production-accessible
- Validated backend reliability using **unittest**, achieving consistent test coverage across core modules

Shipping Document Validator System – Django, OCR

- Developed an OCR-based system to automate document validation
- Handled backend processing and improved error handling using logs
- Reduced manual verification effort through automation

Railway Stock Management System – Django, PostgreSQL

- Built an inventory management system for tracking stock and warehouse data
- Managed database operations and ensured data consistency using SQL queries
- Implemented user roles and basic access control features

AI/ML BASED PROJECTS

Weapon Detection System – Python, OpenCV, TensorFlow, IoT

- Developed a real-time weapon detection system using Convolutional Neural Networks (CNNs) with TensorFlow, achieving 87% accuracy.
- Optimized the model for edge deployment on IoT-based surveillance cameras for low-latency threat detection.

AI Automation Platform – Selenium, PyQt5, OpenCV

- Built a Python-based automation platform integrating web scraping, API testing, and GUI automation.
- Designed a user-friendly PyQt5 desktop interface enabling non-technical users to perform automation tasks efficiently.

Smart QR Attendance System – Python, MySQL, OpenCV, NFC

- Developed a multi-modal attendance system using QR codes, NFC, and facial recognition with 95% accuracy.
- Implemented real-time database synchronization using MySQL for efficient attendance tracking and reporting.

PROFESSIONAL EXPERIENCE

Full Stack Developer Intern – Vdart Academy

1 Month

- Developed the **Shipping Document Validator System** using **Django and OCR**, automating document verification workflows
- Built backend logic and handled API integrations for document processing and validation
- Worked with **PostgreSQL/MySQL** for storing and managing document data
- Debugged issues, improved error handling, and ensured smooth application functionality

Python Developer Intern (Remote) – Outrix

1 Month

- Developed multiple Python-based applications including a **Weather App, Scientific Calculator, Map API integration app, and utility tools**
- Implemented API integrations and handled data processing for real-time applications
- Focused on writing clean, efficient code and handling edge cases for reliable performance
- Gained hands-on experience in building small-scale real-world applications and debugging

CERTIFICATIONS

Django Web Development – Guvi (IIT Madras)

Python Selenium Automation – Guvi (IIT Madras)

Web Scraping – Great Learning

CCNT (Cisco Certified Network Technician) – Cisco

Docker For Beginners -KodeKloud

TECHNICAL EVENTS & COMPETITIONS

- Participated in **National Technical Exhibition** at Dr. Ambedkar Institute, Bangalore, showcasing technical project work
- Participated in **CodeHunters – Algorithm Optimization Challenge**, focusing on problem-solving and logical thinking
- Participated in **HackSprint25 – 24-Hour Hackathon**, collaborating in a team to develop a working web-based solution under time constraints

Competitive Achievements:

- Advanced to **Round 3 of TCS CodeVita**, demonstrating strong algorithmic problem-solving skills
- Qualified for **Round 2 of Zoho Recruitment Process**
- Practises **DSA regularly on LeetCode**, focusing on Arrays, Strings, HashMaps, and Dynamic Programming